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Architecting a Web Platform for Collaborative Documentation in Digital Fabrication Design Education

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Abstract: Digital fabrication courses require tools that support iterative, process-oriented documentation and team collaboration. This paper presents the evolution of DesignEDU, a web-based platform for documenting learning progress in university-level digital fabrication education. Across two course iterations involving over 200 students, user feedback systematically drove platform changes: from cloud-hosted to self-hosted infrastructure, from a basic block editor to a feature-rich collaborative editor with real-time collaboration via Conflict-free Replicated Data Types (CRDTs), and from no AI integration to an embedded AI teaching assistant and text improvement capabilities. Surveys in both cohorts confirmed consistent usability alongside persistent critical gaps, namely no autosave, no simultaneous editing, and cloud dependency, that motivated the platform's redesign. We trace each user-identified issue to a specific engineering solution, demonstrating a feedback-driven methodology for evolving educational technology in design education contexts.

Keywords: digital fabrication, collaborative documentation, educational technology, user feedback, web platform

1 Introduction

Digital fabrication education requires students to document iterative design processes, including failures, experiments, and collaborative decisions, as an integral part of learning (Milara et al., 2019; Georgiev et al., 2023). Unlike traditional coursework, digital fabrication projects unfold in makerspaces and FabLabs where physical prototyping, team coordination, and process reflection are central activities. Existing documentation tools, while valuable, tend to focus on specific aspects: capturing prototype images (Barhoush et al., 2019), enabling effortless capture of design output for building design repositories (Sjöman et al., 2017), digitally capturing physical prototypes for analysis of project progression (Erichsen et al., 2021), recording making steps (Tseng et al., 2014), or supporting multimodal documentation (Gourlet et al., 2016). Few platforms address the full lifecycle of documentation, collaboration, assessment, and feedback within a single integrated environment.

To address this gap, we developed DesignEDU, a web platform designed specifically for digital fabrication courses at the University of Oulu. DesignEDU was first deployed in 2024 to support 128 students across 32 teams in the Principles of Digital Fabrication course (5 ECTS). An initial evaluation (Fernando et al., 2025) described the platform's architecture and evaluated AI-powered summarisation of student submissions. That work focused on the technical foundation and AI model comparison.

This paper presents the subsequent evolution of DesignEDU, driven by systematic feedback from students and teachers collected during two deployments (2024, 115 of 128 students; 2025, 111 of 120 students). The contribution is distinct from prior work: rather than describing a static system, we trace how specific user-identified issues, confirmed across two independent cohorts, and institutional requirements shaped concrete platform changes. These include migrating from cloud-hosted to self-hosted infrastructure, replacing the text editor to support real-time collaborative editing, and integrating AI services for teaching assistance and text improvement.

Our research question is: *How can systematic user feedback drive the iterative evolution of a web platform for collaborative documentation in digital fabrication design education?*

This work contributes to the understanding of socio-technical systems by demonstrating how technology development must respond to the interplay of pedagogical needs, institutional constraints, and student expectations. The platform's evolution illustrates value-based design principles, namely data ownership and control, collaboration equity, and privacy-preserving AI, embedded in engineering decisions.

2 Background and Related Work

2.1 Documentation Tools in Design Education

Several systems have been developed to support documentation in design and fabrication education. Build in Progress (BiP) (Tseng et al., 2014) is a web-based tool using a branching tree structure to visualise design processes. While effective for showcasing project scope, BiP focuses more on recreating final products than reflecting on intermediate stages. Määttä and Troxler (2011) developed Fab Lab documentation prototypes, including a Drupal-based kiosk using a "check-in/check-out" method and a WordPress blog for content sharing. They also proposed FabML, an XML-based format for standardising documentation across labs.

Milara et al. (2019) created "Document-While-Doing," a mobile application that encourages in-progress documentation by automatically capturing and uploading images for annotation. This system promotes process-oriented documentation, including failures and problems. Protobooth Oulu (Barhoush et al., 2019) automates physical prototype capture using multiple cameras and RFID identification, creating chronological records of prototype development. DoDoc (Gourlet et al., 2016) supports multimodal capture, including photographs, videos, stop-motion, and voice recordings, designed to integrate into creative environments.

While these systems share the goal of documenting design processes, each addresses specific aspects. None provides an integrated web platform combining rich-text documentation, real-time team collaboration, AI-assisted feedback, and course administration within a single environment. DesignEDU was developed to fill this gap.

2.2 Initial DesignEDU Platform (2024)

The initial DesignEDU platform (Fernando, 2024; Fernando et al., 2025) was built with Next.js (a web application framework based on JavaScript) hosted on Vercel (a cloud hosting platform), Convex (a cloud database with real-time data synchronisation), Clerk with Google Single Sign-On (SSO) for authentication, EdgeStore (a cloud-based file storage service) for image management, and BlockNote.js as a Notion-like rich text editor. The platform supported weekly documentation tasks, AI-powered summarisation of student submissions using OpenAI GPT-3.5, email support tickets via Resend (an email delivery service), and an inbuilt survey system. It was deployed for the 2024 iteration of the Principles of Digital Fabrication course, serving 128 students in 32 teams with 4 instructors. The same platform was deployed unchanged for the 2025 iteration (120 students).

2.3 Key Findings from the 2024 and 2025 Deployments

Quantitative evaluation (n=115 survey responses) showed generally positive usability: ease of use scored 3.97/5, learnability 4.09/5, and perceived complexity was low at 2.05/5 (Fernando et al., 2025). However, qualitative feedback revealed critical issues that motivated the platform's redesign:

- **No autosave:** Students reported losing substantial work because the editor required manual saving, the single most frequently reported issue.
- **No real-time collaboration:** Teams could not edit documents simultaneously, creating workflow bottlenecks.
- **Vendor lock-in concerns:** Teachers expressed concerns about dependency on third-party cloud services (Vercel, Convex cloud, EdgeStore) for an educational tool handling student data.
- **Limited mobile experience:** Only 17.97% of students agreed that the mobile interface worked well, with 66.41% neutral.
- **No direct video upload:** Students had to upload videos to YouTube and embed links, adding friction to multimedia documentation.

Teacher feedback emphasised the need for self-hosted deployment on university infrastructure, better navigation between group submissions, structured data management (tagging, word counts, individual contribution tracking), and integration with existing university systems.

These findings were independently confirmed by the 2025 cohort (n=111), where students reported the same top issues, specifically autosave, collaboration, and video upload, despite being an entirely different student population (see Section 4 for detailed comparison). These findings directly shaped the platform evolution described in Section 3.

3 Platform Evolution: Design Rationale and Implementation

This section describes the changes implemented for the 2026 deployment, organised by the user needs they address.

Figure 1 illustrates the architectural evolution.

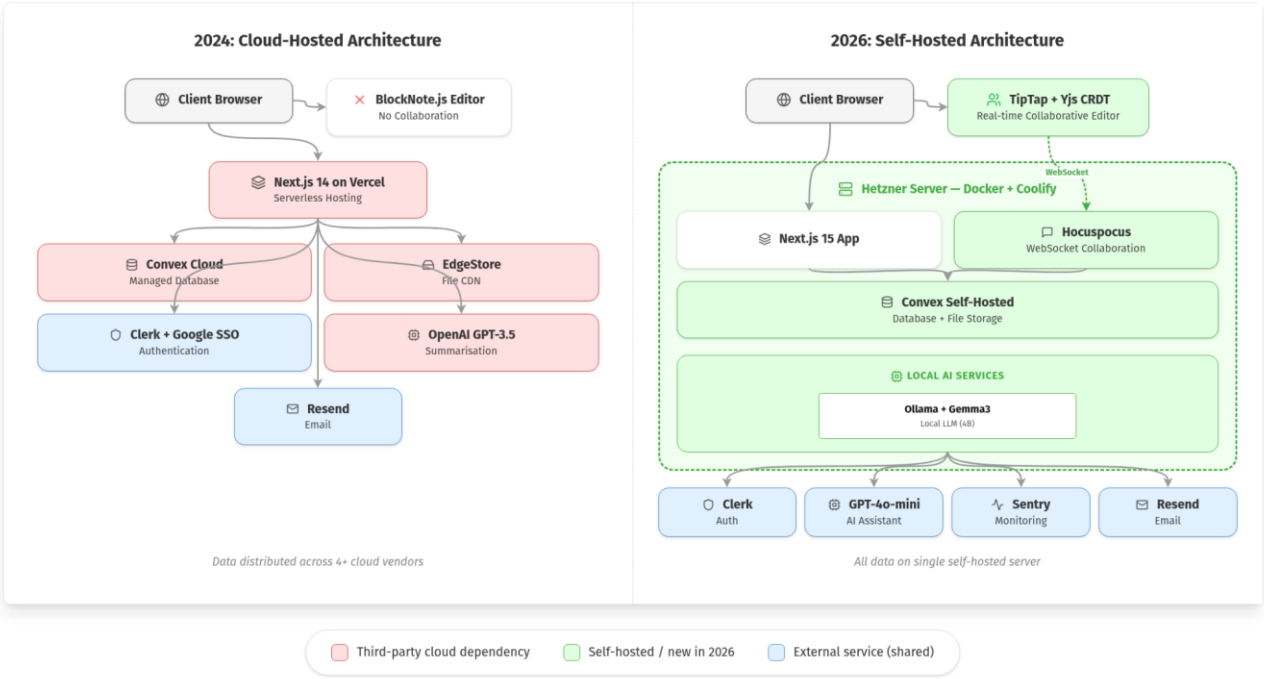


Figure 1. System architecture comparison: 2024 cloud-hosted deployment (left) vs. 2026 self-hosted deployment (right). Red indicates third-party cloud dependencies eliminated in 2026. Green indicates self-hosted components and new capabilities.

The redesign followed an iterative, feedback-driven design process. After the 2024 deployment, we collected structured survey responses and conducted teacher feedback sessions. When the 2025 cohort independently confirmed the same critical issues, specifically the lack of autosave, absence of real-time collaboration, and cloud dependency, we treated these as validated requirements rather than isolated complaints. A key lesson from this process was that single-cohort feedback can reflect cohort-specific preferences, but when an independent cohort confirms the same issues, these become validated requirements warranting architectural changes rather than incremental fixes. Each confirmed issue was mapped to a candidate engineering solution, evaluated against institutional constraints (data ownership, cost, maintainability), and prioritised based on frequency and severity of user reports. This process shaped the specific technical decisions described in the following subsections, ensuring that platform changes were anchored to demonstrated user needs rather than driven by technological trends.

3.1 Infrastructure: From Cloud to Self-Hosted

The most fundamental change was migrating from a fully cloud-dependent architecture to self-hosted infrastructure. Table 1 summarises this transition.

Table 1. Infrastructure evolution from 2024 to 2026

Component	2024 (Cloud)	2026 (Self-Hosted)
Application hosting	Vercel (serverless)	Hetzner server + Coolify
Database	Convex cloud (managed)	Convex self-hosted (Docker)
File storage	EdgeStore (3rd party CDN)	Convex Storage (self-hosted)
Deployment	Vercel auto-deploy	Git push → Coolify auto-deploy
AI services	OpenAI API (cloud only)	OpenAI API + Ollama (local AI runner)
Data location	Distributed across vendors	Single server, full control
Cost model	Free tiers with usage limits	Fixed monthly server cost

This migration was driven primarily by teacher feedback requesting deployment on infrastructure the university could control. Specific concerns included full data ownership and control for student submissions, avoiding vendor lock-in with free-tier services that could change pricing, and the ability to customise the deployment environment. The self-hosted Convex instance runs as a Docker container (a lightweight, portable software package that includes everything needed to run an application) on a Hetzner dedicated server, with Coolify providing a management interface for automated deployments. This approach preserves Convex's real-time synchronisation capabilities while eliminating dependency on the cloud-managed service.

3.2 Editor Migration: BlockNote.js to TipTap

The text editor was replaced from BlockNote.js to TipTap to address both the autosave problem and the need for richer editing features. Table 2 compares the two editors as configured in DesignEDU.

Table 2. Editor feature comparison

Feature	BlockNote.js (2024)	TipTap (2026)
Tables	Not supported	Full table support with operations
Code blocks	Basic	Syntax highlighting (Python, JS, CSS, HTML, JSON, Bash)
Math equations	Not supported	KaTeX integration
Resizable images	Not supported	Drag-and-drop with resize handles
YouTube embeds	Not supported	Native embed support
Task lists	Basic	Checklists with completion tracking
Slash commands	Basic	Extended with AI actions
Real-time collaboration	Not compatible	Native Yjs/Hocuspocus integration
AI text improvement	Not available	Integrated via API endpoint
Emoji picker	Not available	Built-in picker

The migration to TipTap was motivated by two factors. First, TipTap's architecture natively supports Yjs-based collaboration, enabling the real-time editing that students had requested. Second, the richer feature set (tables, math, code highlighting) better serves the technical documentation needs of digital fabrication projects, where students document code, calculations, and structured data.

3.3 Real-Time Collaborative Editing

The absence of real-time collaboration was the most impactful issue identified in the 2024 deployment. Students working in teams of 3-4 could not edit documents simultaneously, forcing sequential workflows and causing content conflicts.

Traditional approaches to real-time collaboration relied on Operational Transformation (OT) (Sun & Ellis, 1998), which requires a central server to coordinate edits. CRDTs offer a decentralised alternative where concurrent edits converge automatically without coordination (Kleppmann & Beresford, 2017). The 2026 platform implements real-time collaborative editing using three CRDT-based components:

- **Hocuspocus** (v3.4.0):A WebSocket server (a technology enabling persistent two-way communication between users' browsers and the server) that manages document synchronisation between connected clients.
- **Yjs** (v13.6.27): A CRDT (Conflict-free Replicated Data Type) library (Shapiro et al., 2011) that enables automatic conflict resolution when multiple users edit concurrently.
- **TipTap Collaboration extensions** (v2.11.3): Editor plugins that connect TipTap to the Yjs document model.

This architecture provides Google Docs-style simultaneous editing with live cursor positions showing each user's name and colour, automatic conflict resolution without manual merging, connection status indicators showing online/offline state, and offline support with automatic synchronisation on reconnect. The collaboration server runs on the same Hetzner infrastructure, keeping all synchronisation data within the self-hosted environment.

3.4 AI Integration

The 2026 platform integrates three AI services, each serving a distinct purpose. This work is distinct from the AI summarisation evaluation reported in Fernando et al. (2025), which compared language models for automated summary generation. The current integrations focus on interactive assistance:

AI Teaching Assistant. A chatbot powered by OpenAI GPT-4o-mini provides course-context-aware responses to student queries. The assistant is configured with knowledge of the Digital Fabrication curriculum and can guide students on assignments, project analysis, and fabrication techniques. It uses server-side streaming for real-time responses and stores conversation history in the Convex database for continuity.

Local LLM for Text Improvement. Ollama with the Gemma3 model (4B parameters, ~3 GB) runs locally on the Hetzner server for AI-assisted text improvement within the editor. Students can select text and request improvements via a slash command, with processing handled entirely on the server without external service calls. This addresses privacy concerns while providing AI writing assistance at no marginal cost.

3.5 New Administrative Features

Several administrative features were added in response to teacher feedback:

- **Attendance tracking:** Bulk CSV import from spreadsheets with per-week tracking (weeks 2–8), duplicate detection, and multiple query methods (by group, student, or week).
- **Notification system:** Teachers can send targeted notifications to specific groups with survey links, deadline reminders, and task modification alerts.
- **Team project submissions:** A structured submission form requiring images of unique features, teamwork process, inner workings, and isometric views, with metadata tracking and admin dashboard.
- **PDF generation:** Automated generation of formatted PDF documents from submissions for archival and review.
- **SurveyJS integration:** Replaced the custom inbuilt survey with the SurveyJS library, supporting complex question types, branching logic, and structured response collection.
- **Error monitoring:** Sentry integration for tracking runtime errors and performance issues in production.

3.6 File Storage Migration

File storage was migrated from EdgeStore (a third-party service) to Convex Storage (self-hosted). Table 3 summarises this change.

Table 3. File storage migration

Aspect	EdgeStore (2024)	Convex Storage (2026)
Storage location	External cloud content delivery network	Self-hosted server volume
URL generation	Permanent public URLs	Dynamic signed URLs
Integration	Separate software library	Native Convex API
Cost	Separate billing (1.5 GB free tier)	Included in server hosting
Data control	Third-party managed	Full institutional control
File limit	Free tier constraints	Server disk capacity

The migration consolidated all data, including documents, user records, and uploaded files, within the self-hosted infrastructure, fulfilling the teacher requirement for full data ownership and control.

4 Evaluation

4.1 Quantitative Analysis (2024 and 2025 Surveys)

Mandatory surveys were administered at the end of each course iteration: 2024 (115 of 128 students, 89.8% response rate) and 2025 (111 of 120 students, 92.5% response rate). Both surveys used identical 5-point Likert scales (1=Strongly Disagree to 5=Strongly Agree) adapted from the System Usability Scale. The 2025 iteration used the same platform as 2024 (no major changes between these years); the redesigned platform described in Section 3 was deployed for the 2026 iteration. Table 4 presents the comparative results.

Table 4. Usability survey results across two cohorts (2024: n=115, 2025: n=111)

Metric	2024 Mean	2024 SD	2025 Mean	2025 SD	Change
Ease of use	3.97	1.00	4.03	1.00	+0.06
Perceived complexity (R)	2.05	0.89	2.16	1.04	+0.11
Learnability	4.09	0.85	4.09	0.97	0.00
Learning curve (R)	1.90	0.91	2.07	1.11	+0.17
Need for assistance (R)	1.79	1.02	2.01	1.28	+0.22
Confidence in use	3.81	0.98	3.86	1.01	+0.05
Design consistency	3.88	0.78	3.95	1.04	+0.07
Feature integration	3.58	0.86	3.67	1.07	+0.09
Relevance to learning	3.07	0.94	3.42	1.09	**+0.35**
Mobile usability	3.20	0.68	3.28	0.92	+0.08
Ease vs. other LMS	3.56	1.01	3.41	1.25	-0.15
Feature usefulness	3.55	0.69	3.62	0.91	+0.07

(R) = negatively worded item; lower values indicate better usability for these items

The results show remarkable consistency across two independent cohorts. Core usability strengths persisted: learnability remained the highest-rated item at 4.09/5, ease of use improved slightly to 4.03/5, and the majority of students in both years reported not needing assistance. The most notable change was relevance to learning, which improved from 3.07 to 3.42 (+0.35), with agreement rising from 26.56% to 44.1%, suggesting the platform's educational value was increasingly recognised.

Reverse-scored items showed small unfavourable shifts (complexity +0.11, need for assistance +0.22, learning curve +0.17), possibly reflecting an expanded feature set that added perceived complexity. Mobile usability remained the weakest area, with only 27.0% of 2025 students agreeing the mobile interface worked well (up from 17.97% in 2024) and 65.8% remaining neutral.

Assignment submission behaviour in 2024 improved over the course duration: on-time submissions rose from 50.0% in Week 1 to 96.9% in Week 8, while average late submission delays decreased from 12 days 16 hours to 1 day 16 hours. This pattern suggests increasing platform familiarity and engagement, though course-level factors (approaching final deadlines) may also contribute.

4.2 Qualitative Feedback Analysis

Open-ended survey responses from both cohorts and structured teacher feedback interviews identified consistent strengths and critical issues across two independent student populations.

Student-reported strengths (both years) included easy project documentation with a clean interface, smooth and fast navigation, secure Google SSO login, and good multimedia embedding support. Students noted that the platform positively impacted project management and meeting deadlines. In 2025, students described the platform as "simple to use" and the documentation editor as "pretty handy and easy to use."

Student-reported issues were remarkably consistent across both cohorts, centring on five persistent themes: (1) loss of work due to no autosave, the single most prevalent issue in both years, with 9 explicit requests in 2025 alone, including one student who wrote: "Lack of autosaving makes this platform terrible. I wrote weekly report directly to this platform and accidentally pressed previous page hotkey from mouse"; (2) inability to collaborate simultaneously, with 2025 students reporting that "if two people save at the same time it sometimes deletes everything"; (3) Google account login friction, the most cited difficulty in 2025 (8+ mentions), with students finding the multi-step university Google account activation process frustrating; (4) dependence on external YouTube links for video content (5 mentions in 2025); and (5) limited mobile responsiveness.

Teacher feedback emphasised institutional concerns: deployment within university infrastructure using Docker for data control, integration with existing systems (Moodle, Microsoft 365), structured submission formats ensuring completeness, tools for tracking individual student contributions within team work, and notification systems for deadline management.

The consistency of reported issues across two independent cohorts (2024 and 2025) strengthens the evidence that these were genuine, persistent usability issues rather than cohort-specific artefacts.

4.3 Mapping Feedback to Platform Evolution

Table 5 presents the traceability from identified issues to implemented solutions, demonstrating the feedback-driven development approach.

Table 5. Feedback-to-feature traceability matrix

Issue Identified	Source	Confirmed in 2025?	Solution Implemented (2026)
No autosave; work lost	Students	Yes (9 mentions, #1 request)	TipTap + Yjs with continuous autosave via CRDT sync
No real-time collaboration	Students	Yes (4 requests + 5 conflict reports)	Hocuspocus WebSocket + Yjs collaborative editing
Vendor lock-in concerns	Teachers	N/A	Self-hosted on Hetzner + Coolify + Docker
Third-party file storage dependency	Teachers	N/A	Convex Storage (self-hosted) replacing EdgeStore
No direct video upload	Students	Yes (5 mentions)	Convex Storage file uploads (3 MB limit)
Limited editor features	Students	Yes (image paste, checkboxes)	TipTap with tables, code blocks, math, embeds
No notification system	Teachers	Yes (1 mention)	Group notification system with targeted messaging
No attendance tracking	Teachers	N/A	Bulk CSV import with per-week tracking
No structured submissions	Teachers	N/A	Team submission forms with structured image uploads
Custom survey limitations	Teachers	N/A	SurveyJS integration with complex question types
No AI assistance for students	Teachers	N/A	GPT-4o-mini chatbot + local Gemma3 text improvement
Cloud-only AI processing	Teachers	N/A	Local Ollama + Gemma3 for privacy-preserving AI
Login friction (Google SSO)	Students	Yes (8+ mentions, #1 difficulty)	Under consideration (Microsoft 365 integration)

This traceability demonstrates that every major platform change originated from specific user needs identified during the 2024 and 2025 deployments, rather than from technology-driven decisions, consistent with evidence that systematic requirements tracing improves software quality (Lucassen et al., 2016). The persistence of the same top issues across two independent cohorts (autosave: #1 in both years; collaboration conflicts: top 3 in both years; video upload: requested in both years) provides strong validation that these were genuine, systemic issues requiring architectural solutions.

4.4 Platform Scale and Growth

The platform has grown substantially between deployments. Both the 2024 and 2025 iterations used the same platform architecture (cloud-hosted, no real-time collaboration); the redesigned architecture described in Section 3 is deployed for the 2026 iteration. Table 6 summarises the platform's scale across both deployments.

Table 6. Platform scale across deployments

Metric	2024	2025
Students	128	120
Teams	32	32
Instructors	4	4
Documentation pages	256	274
Survey responses	115	111
AI-generated summaries	18	16
Total database records	836	898

Figure 2 shows screenshots of the key new features introduced in the 2026 redesign.

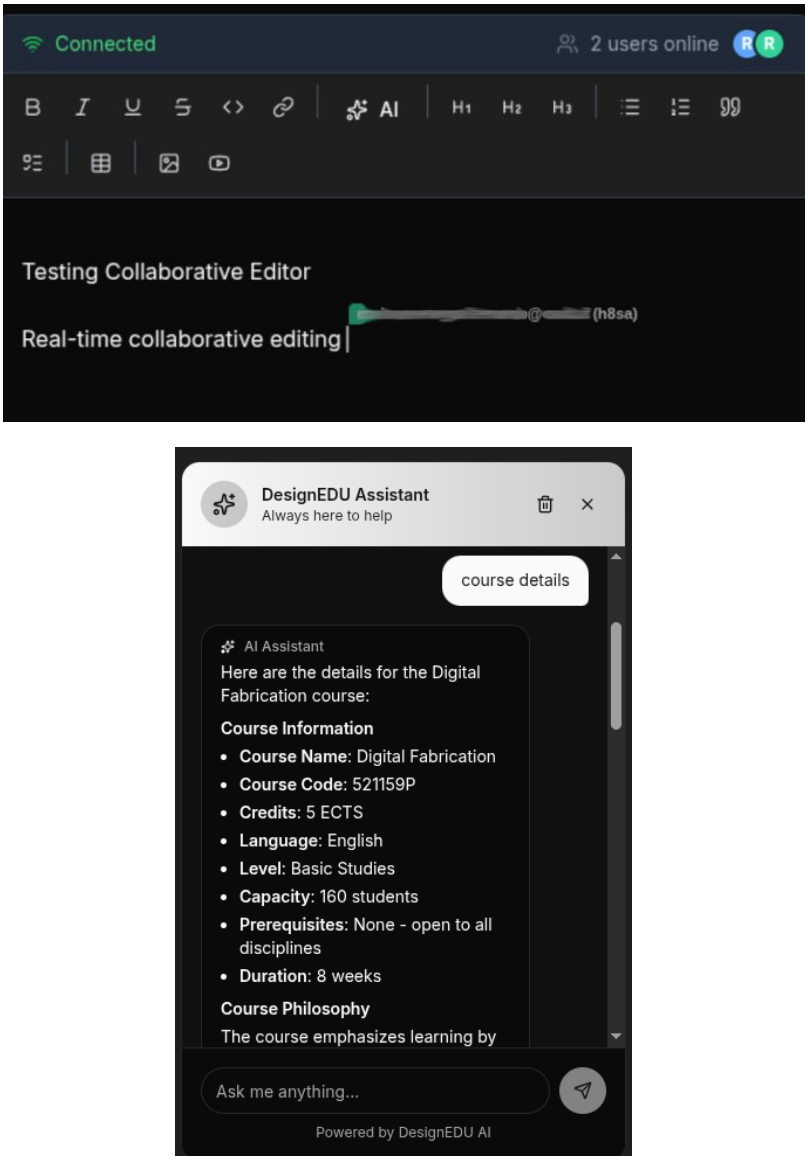


Figure 2. Screenshots of key new features: (top) real-time collaborative editing; (bottom) AI teaching assistant

5 Discussion

5.1 Feedback-Driven Development as Methodology

This work demonstrates that systematic collection and tracing of user feedback provides a principled methodology for evolving educational platforms. The traceability matrix (Table 5) shows that each significant engineering decision, whether self-hosting, editor migration, or AI integration, was anchored to specific user needs rather than driven by technological novelty. The confirmation of the same issues across two independent cohorts (n=226 total) strengthens this evidence: when two different student populations independently identify the same top issues (autosave, collaboration, video upload), these represent genuine systemic needs rather than cohort-specific preferences. This approach is particularly relevant for educational technology, where the diversity of stakeholders (students, teachers, administrators) generates requirements that may conflict or prioritise differently.

5.2 Socio-Technical Alignment

The platform evolution illustrates the socio-technical nature of educational technology design. The migration to self-hosted infrastructure was not a purely technical decision but reflected institutional values around full data ownership and control and long-term sustainability. Similarly, the choice to run AI services locally (Gemma3) alongside cloud APIs (GPT-4o-mini) represents a deliberate balance between capability and privacy. These decisions exemplify value-based design, where engineering choices embed and reflect organisational values.

5.3 Trade-Offs

Self-hosting introduces operational complexity: server maintenance, backup management, security updates, and capacity planning become the institution's responsibility rather than the cloud provider's. For the DesignEDU deployment, this trade-off was acceptable because the course runs on a predictable annual cycle, the student population is bounded (~160 students), and the instructional team includes members with systems administration expertise. This trade-off calculus may differ for institutions without such expertise.

The local AI model (Gemma3 4B) prioritises privacy and cost elimination over capability. The Gemma3 model at 4 billion parameters provides adequate text improvement suggestions for student documentation but cannot match the reasoning capabilities of larger cloud-hosted models. This represents a conscious design choice: for routine text improvement within an educational context, adequate local performance outweighs superior cloud performance when privacy and cost are considered.

5.4 Comparison with Existing Work

Our findings both extend and diverge from prior work on documentation tools in design education. Tseng et al. (2014) identified the challenge of representing design processes online, noting that existing tools emphasised final outcomes over intermediate stages. DesignEDU addresses this gap by structuring documentation around weekly iterative entries, and our survey data confirms that students valued this process-oriented approach (relevance to learning improved from 3.07 to 3.42/5 across cohorts). This extends Tseng et al.'s observation by demonstrating that structured weekly documentation can increase perceived educational value over successive course iterations.

The effortless capture principle advocated by Sjöman et al. (2017) finds direct support in our results. The most persistent user-identified issue across both cohorts was the lack of autosave, with students losing work due to manual saving requirements. This confirms that reducing friction in the documentation process is not merely a convenience but a critical usability factor, consistent with Sjöman et al.'s argument that effortless capture is a prerequisite for meaningful design documentation.

While hardware-based approaches such as Protobooth (Barhoush et al., 2019; Erichsen et al., 2021) and Document-While-Doing (Milara et al., 2019) focus on capturing physical artefacts, our work addresses the complementary challenge of integrating text-based documentation, team collaboration, and AI assistance within a single software platform. Unlike these systems, which target specific documentation activities, DesignEDU combines multiple functions (documentation, assessment, feedback, administration) in one environment. Our evaluation suggests that this integration is valued by users (feature integration scored 3.58–3.67/5), though it introduces complexity that some students noted.

Regarding AI in education, Zawacki-Richter et al. (2019) noted the scarcity of educator-facing AI applications in higher education. Our approach of embedding AI directly within the documentation workflow, rather than as a separate tool, represents a practical response to this gap. However, our finding that AI-generated summaries decreased from 18 (2024) to 16 (2025) despite comparable student numbers suggests that AI summarisation features may see limited organic adoption without stronger integration into grading workflows, an observation not well addressed in the existing literature.

The self-hosting decision aligns with the "local-first" software principles articulated by Kleppmann et al. (2019), who argued that users should retain ownership and control of their data rather than depending on cloud services. Our experience confirms that self-hosting provides greater institutional control but requires sustained technical expertise, a trade-off that our bounded course context (predictable annual cycle, ~160 students) made feasible.

5.5 Limitations

This study is conducted at a single institution within a specific course context (introductory digital fabrication). The findings may not generalise to other educational domains, course structures, or institutional contexts. The quantitative evaluation data is from the 2024 and 2025 deployments, both using the pre-redesign platform; systematic evaluation of the 2026 platform with the new features is planned for the current course iteration but is not yet available. While the two-cohort consistency strengthens confidence in the identified issues, a direct before/after comparison of the redesigned platform remains future work. Additionally, the self-hosting approach requires technical expertise that may not be available at all institutions.

5.6 Future Work

Future directions include cross-institutional deployment to test generalisability, longitudinal studies tracking how platform usage patterns evolve across multiple course iterations, integration of more advanced AI features such as personalised learning paths and automated project assessment, and systematic comparison of learning outcomes between the 2024 and 2026 platform versions.

6 Conclusion

This paper presented the feedback-driven evolution of DesignEDU, a web platform for collaborative documentation in digital fabrication design education. Through systematic collection and analysis of student and teacher feedback across two cohorts (2024: n=115, 2025: n=111), we identified critical gaps, including missing autosave, no real-time collaboration, cloud dependency, and limited AI integration, confirmed their persistence across independent student populations, and traced each to specific engineering solutions in the 2026 redesign.

The key contributions are: (1) a traceable methodology linking user feedback to platform design decisions in educational technology; (2) a concrete case study of migrating an educational platform from cloud-hosted to self-hosted infrastructure, balancing full data ownership and control against operational complexity; and (3) an integrated architecture combining real-time collaborative editing, privacy-preserving local AI services, and course administration within a single platform serving 160+ concurrent students.

DesignEDU demonstrates that educational platforms benefit from iterative, user-centred development where technology choices are anchored to demonstrated needs rather than technological trends. This approach aligns with value-based design principles and contributes to the broader understanding of socio-technical systems in design education.

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